UNIVERSITY DIGITAL CLEARANCE SYSTEM (Unipass) – PROJECT DOCUMENTATION

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1. Introduction

Project Overview

The **University Digital Clearance System (UDCS)** is a web-based platform designed to digitize and automate the student clearance process before and after graduation.

It replaces the traditional paper-based clearance forms that require physical signatures from multiple departments (library, faculty, bursar, dean, etc.), which often leads to:

- Delays due to unavailable staff
- Congestion in offices
- Loss of forms
- Forgery of signatures
- Corruption (especially for urgent transcript certification)

The system ensures a **secure, fast, and transparent clearance process**.

Objectives

- Eliminate paper-based clearance forms
- Reduce delays caused by unavailable staff
- Prevent forgery through digital verification
- Minimize corruption via transparent workflows
- Enable students to complete clearance **within minutes instead of days**

Target Users

- Students
- Department Officers

- Faculty Dean
- Dean of Students
- Bursar
- Academic Registrar (AR)
- System Administrators

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2. Problem Statement

Current manual clearance system issues:

- Students move physically across offices
- Officers may be absent or busy
- Forms are signed without proper verification
- Some signatures are forged
- Students pay unofficial fees for faster services

Thus resulting into ineffenciency,unfaireness in signature delivery and unaccountability

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3. Proposed Solution

A **web-based clearance system** that:

- Assigns clearance tasks digitally to each department
- Tracks progress in real time
- Uses **secure authentication instead of signatures**
- Allows officers to approve/reject clearance online
- Logs all actions to prevent fraud

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4. System Requirements

Functional Requirements

4 Student Module

- Students log in using university email
- View clearance status (all departments)

- Submit clearance request
- Receive notifications on approval/rejection

Department Clearance Module

Each department (Library, Faculty, Bursar, etc.):

- Views assigned students
- Approves or rejects clearance
- Adds comments (e.g., “Return borrowed book”)

Clearance Workflow

Clearance follows this order:

1. Department(s)
2. Library
3. Faculty Dean
4. Dean of Students
5. Bursar
6. Academic Registrar (Final Approval)

Admin Module

- Manage users and roles
- Monitor clearance progress
- View logs and reports
- Configure departments

Transcript Request Module

Students request certified transcripts

- System logs request time
- Officers process requests transparently
- No manual shortcuts allowed

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5. Non-Functional Requirements

Usability

- Simple dashboard with status indicators (✓ Approved / ✕ Pending)
- Students complete process **within 3 minutes**
- Minimal navigation (no more than 3 clicks per action)

Performance

- System loads within **8 seconds**
- Dashboard loads within **3 seconds**
- Efficient database queries

Security

- Role-based access control
- Password hashing (bcrypt)
- No manual signatures (digital approval only)
- Unique login credentials for each officer
- Activity logging (who approved what and when)

Transparency and Anti-Corruption

- Timestamp for every approval
- No bypassing steps
- Audit logs visible to admin
- No hidden approvals

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6. System Architecture

Technology Stack

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** PHP
- **Database:** MySQL
- **Server:** WAMP/XAMPP

Architecture Design

- Client Browser (User Interface)
- PHP Backend (Business Logic)
- MySQL Database (Data Storage)

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7. System Modules

Authentication Module

- Email-based login
- Role detection (Student, Officer, Admin)
- Secure session handling

Clearance Management Module

- Tracks clearance status per department
- Displays progress bar
- Prevents skipping steps

Approval Module

- Officers approve/reject requests
- Add comments
- Digital record stored

Notification Module

- Alerts for:

    - Approval
    - Rejection
    - Pending action

Audit and Logging Module

- Logs:
    - Login attempts
    - Approvals
    - Rejections
- Prevents forgery and corruption

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8. Database Design

Main Tables

- Users (id, email, password, role)
- Departments (id, name)
- Clearance Requests (id, student_id, status)
- ClearanceSteps (id, request_id, department_id, status, comment)
- Logs (id, user_id, action, timestamp)
- TranscriptRequests (id, student_id, status)

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9. User Interface Design

Student Dashboard

- Clearance progress bar
- Department status list
- Submit request button

Officer Dashboard

- List of students to clear
- Approve/Reject buttons

- Comment field

Admin Dashboard

- Total students cleared
- Pending requests
- Logs and reports

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10. System Workflow

1. Student logs in
2. Submits clearance request
3. System assigns departments
4. Each officer approves/rejects
5. Status updated in real-time
6. Final approval by Academic Registrar
7. Clearance completed

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11. Key Innovations

No Physical Movement

- Entire process online

Anti-Forgery System

- No handwritten signatures
- Digital approvals with user accounts

Anti-Corruption Measures

- Transparent logs
- No hidden processing
- Equal processing order

Bulk Processing (Class Clearance)

- Admin can upload class list
- Officers can clear multiple students at once

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12. Performance Optimization

- Optimized SQL queries
- Indexed database tables
- Minimal page reload
- Compressed assets

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13. Testing Strategy

Test Cases

- Student submits request → Visible to all departments
- Officer approves → Status updates correctly
- Unauthorized user → Access denied
- Skipping steps → Not allowed

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14. Future Enhancements

- Mobile App
- SMS notifications
- Integration with student records system
- Online payment for transcripts (mobile money)

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15. Conclusion

The Digital Clearance System solves major challenges in the traditional process by:

- Eliminating delays
- Preventing forgery
- Reducing corruption
- Improving efficiency

It ensures a **fast, secure, and transparent clearance process** suitable for modern universities.